

REFERENCES.md

SpeechStudio Reference Documentation

Version: 1.0

Status: Approved

1. Purpose

This document lists every official and approved reference used during the development of SpeechStudio.

Only sources listed in this document may be considered authoritative.

If two sources conflict, priority is determined by the order defined below.

2. Source Priority

Priority 1

Official BosonAI documentation

Priority 2

Official Hugging Face model documentation

Priority 3

Official BosonAI GitHub repository

Priority 4

Official implementation examples

Priority 5

SpeechStudio verified experiments

Priority 6

SpeechStudio Deep Research documents

Information from lower priority sources must never override higher priority sources without verification.

3. Official Model

Model

BosonAI Higgs Audio V3 TTS 4B

Repository

<https://huggingface.co/bosonai/higgs-tts-v3-4b>

Purpose

Primary speech synthesis model.

Status

Official.

4. Transformers Implementation

Model

multimodalart Higgs Audio V3 Transformers Port

Repository

<https://huggingface.co/multimodalart/higgs-audio-v3-tts-4b-transformers>

Purpose

Current backend used by SpeechStudio.

Status

Approved implementation.

5. Official Prompt Documentation

Document

PROMPTING.md

Repository

<https://huggingface.co/bosonai/higgs-tts-v3-4b/blob/main/PROMPTING.md>

Purpose

Official prompt syntax.

Emotion tokens.

Style tokens.

Prosody tokens.

Sound effect tokens.

Prompt construction.

6. Official GitHub Repository

Repository

<https://github.com/boson-ai/higgs-audio>

Purpose

Reference implementation.

Examples.

Model loading.

Backend architecture.

7. SGLang Cookbook

Document

Higgs Audio Cookbook

Repository

https://sgl-project.github.io/sglang-omni/cookbook/higgs_tts.html

Purpose

Reference implementation.

Performance notes.

Prompt examples.

API usage.

Only implementation details applicable to the local Transformers backend shall be adopted.

8. vLLM Omni Documentation

Document

BosonAI Higgs Audio V3 TTS

Repository

<https://github.com/vllm-project/vllm-omni/blob/main/recipes/BosonAI/Higgs-Audio-V3-TTS.md>

Purpose

Generation pipeline reference.

Configuration reference.

Performance reference.

Only implementation concepts applicable to SpeechStudio shall be used.

9. Benchmark Reference

Document

benchmark_tts_seedtts.py

Repository

https://github.com/sgl-project/sglang-omni/blob/main/benchmarks/eval/benchmark_tts_seedtts.py

Purpose

Benchmark methodology.

Performance comparison.

Generation workflow reference.

10. SpeechStudio Research

Internal documents.

Approved as project references.

Research documents include

Higgs prompting research

Generation parameter research

Voice cloning research

Performance research

UI research

These documents are stored in

research/

11. Verified Experiments

SpeechStudio maintains its own verified test results.

Only reproducible experiments are considered valid references.

Examples

Emotion tests

Style tests

Prosody tests

Sound effect tests

Voice cloning tests

Generation parameter tests

Silence padding tests

Performance measurements

These documents are stored in

research/tests/

12. Generation Parameters

Official parameter definitions are taken from

Official Hugging Face documentation

Verified implementation

SpeechStudio experiments

Only parameters verified through these sources shall appear in the UI.

13. Prompt Tokens

Prompt syntax is derived from

PROMPTING.md

SpeechStudio verification

Every supported token used by SpeechStudio must exist in the official documentation or be experimentally verified.

14. Voice Cloning

Reference voice recommendations are based on

Official documentation

SpeechStudio experiments

Reference audio validation rules

Reference transcript handling

Reference sample rate

Reference duration

15. Performance

Performance recommendations are based on

Official documentation

CUDA behaviour

SpeechStudio benchmarks

Measured generation times

Measured VRAM usage

Measured realtime factor

16. Internal Specifications

The following project documents are normative.

00_PROJECT.md

01_REQUIREMENTS.md

02_BOOTSTRAP.md

03_ARCHITECTURE.md

04_UI.md

05_PROMPT_SYSTEM.md

06_GENERATION_ENGINE.md

07_VOICE_SYSTEM.md

08_SETTINGS.md

09_STORAGE.md

10_AI_DEVELOPMENT_RULES.md

These documents define SpeechStudio behaviour.

17. External Libraries

SpeechStudio currently depends on

Python

PyTorch

Transformers

Hugging Face Hub

SoundFile

NumPy

Future dependencies require explicit approval.

18. Validation Policy

Information added to SpeechStudio documentation must satisfy at least one of the following

Official BosonAI documentation

Official Hugging Face documentation

Official repository implementation

Verified SpeechStudio experiment

Verified Deep Research finding

Information failing these requirements shall not be treated as specification.

19. Documentation Update Policy

Whenever an official Higgs release changes behaviour

The relevant SpeechStudio specification shall be reviewed.

The change shall be verified experimentally before implementation.

Documentation must be updated before source code.

20. Project Philosophy

SpeechStudio documentation is implementation driven.

Specifications are based on verified behaviour.

Assumptions are not specifications.

Every implementation decision should be traceable to an approved reference or a reproducible experiment.